

Charles Karpati

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Summary of Experience

With 15 years of development experience, 9 of which have been professional, I leverage my extensive software development expertise and deep understanding of system design to create value. My work has significantly impacted various sectors, notably through the creation of an AI email response service valued at over \$10M pre seed with 10,000+ users, development of 3D experiences in leading metaverse platforms, and data analysis for Baltimore city to support the Mayor's data-driven policy decisions. I'm driven by a passion for harnessing technical prowess to create innovative solutions and empower communities. With my documentation and visualization strategies, even the most intricate problems become easily deconstructed and solved.

Work History

Software Developer

03/2024 - 06/2024

AutoApply

- Designed and implemented an AI Chrome Extension to automate form filling. Upload documents, scan webpages, fill forms (React, Sqlite).
- Developed using best practices and guidelines for AI safety to ensure no misuse of the system (JWT, Tokens, OAuth2, Cookies, Sessions).
- Designed backend database infrastructure and file management plan.
- Elicited requirements from customers and assessed their needs to resolve their problems and satisfy expectations.
- Employed machine learning algorithms to develop models and trained and fine-tuned AI large language models and tested them for quality and safety.
- Researched and installed a server rack to self-host private, on-premise cloud services to save costs.
- Registered the domain and managed records for each subdomain to properly host the site.
- Configured SSL, TLS, Certificate Management for each domain.
- Automated DDNS IP updates for the private cloud (Bash, Cloudflare API, Cron) to ensure up-time.
- Re-implemented Sqlite on the browser to share code between server and browser.
- Engineered each server using a suite of installation, startup, and setup scripts (SystemD, .bashrc, Cron, Apache, Ngix, Express).
- Researched and developed a dev-ops pipeline for CI and CD (SFTP, SSH, Cron, DockerCompose).
- Produced and hosted a Latex2PDF service, and an Html2Img, Html2Md, Html2Txt service (PM2, Docker, DockerSwarm, Venv).
- Worked with modern ML best-practices and tooling, like those coming from Langchain, Huggingface, Vertex, and OpenAI. Matplotlib, Numpy, and TensorFlow.

Chief Technology Officer

5/2023 - 2/2024

Addy-Ai

- Co-Founder of AddyAI, an automated AI email response service with over 10000+ users.
- Communicated complex AI concepts to technical and non-technical audiences when marketing our product and raising funds with general public
- Pitched to fortune 500 companies, venture capital funds, and three different mayors.
- Received a \$10 million valuation during the pre-seed fund-raising phase.
- Developed Langdrive, an open-source automated LLM finetuning solution, complete with documentation and tooling to create chatbots and connect to third-party services for document storage and retrieval.
- Identified AI problems, determined relevance, and collaborated with team members to develop recommendations and innovative solutions.
- Evaluated the capabilities, trustworthiness, and safety of these AI models.

- Created Addy's admin panel which enabled AI email customization and third-party integrations.
- Evaluated model safeguards and used real-world adversarial tactics and techniques to identify failures, vulnerabilities, and potential misuses.
- Designed automated diagram generation and code documentation solutions.

Software Developer

3/2022 - 5/2023

Voxels.com

- Lead blockchain developer at a top metaverse company, spearheading web3 initiatives, and fostering relationships with crypto-related business partners (Solidity, Ethers).
- Developed in-game enhancements like real-time video streaming and integration of the "Proof-of-Attendance-Protocol" for in-game events (Babylon, TypeScript).
- Collaborated with UI/UX engineers for site redesigns, and weekly incremental improvements.
- Identified, resolved, and documented a comprehensive list of software bugs.
- Documented system architecture and diagrammed the most complex parts.
- Provisioned and monitored critical infrastructure as needed (DigitalOcean, Heroku).
- Committed to providing quality products and services to our player base.
- Catalogued and resolved most all user-submitted issues with great attention to detail.
- Informed our community of developments, taking into account audience and environment cues.
- Enthusiastically engaged in team discussion, code reviews, and agile methodologies.
- Liaised with MDAO and similar to set NFT metadata standards for cross-metaverse interoperability.
- Offered pro bono metaverse development courses.
- Volunteered as a judge for HackaTRON 2022.

Lecturer

1/2020 - 11/2021

National Science Foundation

- Taught *Python for Big Data* to twenty graduate students from various universities.
- Planned educational curriculum in coordination with the universities.
- Mentored three students through their semester-long data science projects.

Web Developer

1/2017 - 11/2021

Baltimore Neighborhood Indicators Association

- Generated 150 annual quality of life statistics used by Baltimore City's agencies and non-profits.
- Met with the community and key stakeholders to ensure their needs are met and our data is useful.
- Acted as data steward, codified data management policies to enhance data integrity and analytical capabilities.
- Administered the organization's main website and digital presence (Geoloom.com, bnaijfi.org) (WordPress, PHP, JQuery, SQL, Excel, Esri, Socrata).
- Designed Equity tracker, used by the mayor's office to track DEI efforts in the city (Microsoft 365, Power Apps).
- Established Business Integration Search tool, connects consumers to minority-owned businesses (PHP, SQL).
- Engineered VitalSigns, generates BNIA's core data (Ipynb, GeoPandas, Seaborn, Vega, Plotly, MySQL, PhpMyAdmin, PGAdmin, PostgreSQL).
- Developed CloseCrawl, pulls MD Courts data and filters for foreclosures (Python, MechanicalSoup).
- Developed TidyAddr, expertly cleans addresses in Baltimore City (Node).
- Constructed GuidePost, assesses longitudinal trends between communities (JavaScript, Plotly, Vega, Leaflet).
- Assembled BOLD, search property records to study foreclosure trends (React, PHP, SQL, Leaflet).
- Formulated GreenPatterns, identifies greening strategies to reduce watershed runoff (React, PHP, SQL, Leaflet).
- Engineered Dataplay, helps with data intake, cleaning, processing, and visualization.
- Developed Ipynb2web, converts python notebooks into HTML page assets with accompanying audio-visual media.

- Generated FastStat, a chrome extension to quickly retrieve Baltimore City statistics.

Freelance Developer

2009 - Present

- Developed systems for clients including a law firm, an art school, and a major clothing brand.
- Generated Telegram Recap, creates mp3 chat recaps for telegram groups using AI, posted daily.
- Developed NEAR Library, received a grant from the NEAR protocol to build a virtual Library in the metaverse.
- Engineered RideLogger, visualize GPS data captured from my phone using Termux.
- Developed PrintMaps, generate 3D printable boxes in the shape of a map.
- Generated PivotTable, a JavaScript powered Python library for interactive and sync-able data.
- Developed Ipynb2Web, convert python notebooks into assets ready for the web.
- Produced CryptoTrader, an algorithmic trading system - Made 40x returns in one year.
- Developed BT Lights, control an audio-responsive mesh-network of lights.
- Created GPT Watch, an android app for my android watch to ask chatGPT a question.
- Developed Music player, load local mp3 music, YouTube playlists and videos, or explore iTunes.
- Volunteered code to retrieve satellite data for maritime transparency with the Skylight Project.

Education

MS and BS in Information Systems
University of Maryland Baltimore County

Graduated: 2016, 2020

Awards

- GitHub 2024 Open-Source Accelerator alumni - March-June 2024. (AddyAI)
- Venture Bridge Alumni, a Carnegie Mellon accelerator - Jun-Nov 2023. (AddyAI)
- Certified Web Developer, UMBC, 2016

Skills

Node/JavaScript/TypeScript, Python, SQL, NoSQL, HTML, CSS, Bash, PHP, Java and C++, React, Pre-act, Redux, Webpack, Express, Flask, Git, Docker, Heroku, Digital Ocean, WordPress, AWS Lambda, EC2, S3, Google Cloud Console, Bootstrap, Ionic, Tailwind, Styled Components, Google APIs, Cloudflare, Firebase, OAuth2, Twilio, Jira, Esri, Socrata, Leaflet, PostGIS, PostgreSQL, MySQL, SQLite, Redis, GraphQL, Leaflet, D3, Babylon, Three, Matplotlib, Seaborn, Vega, Plotly, Altair, TensorFlow, ONNX, OpenAI, Gradio, HuggingFace, Langchain, Numpy, Pandas, OpenSeas, Solidity, Ethers, Infura, Alchemy, Web3